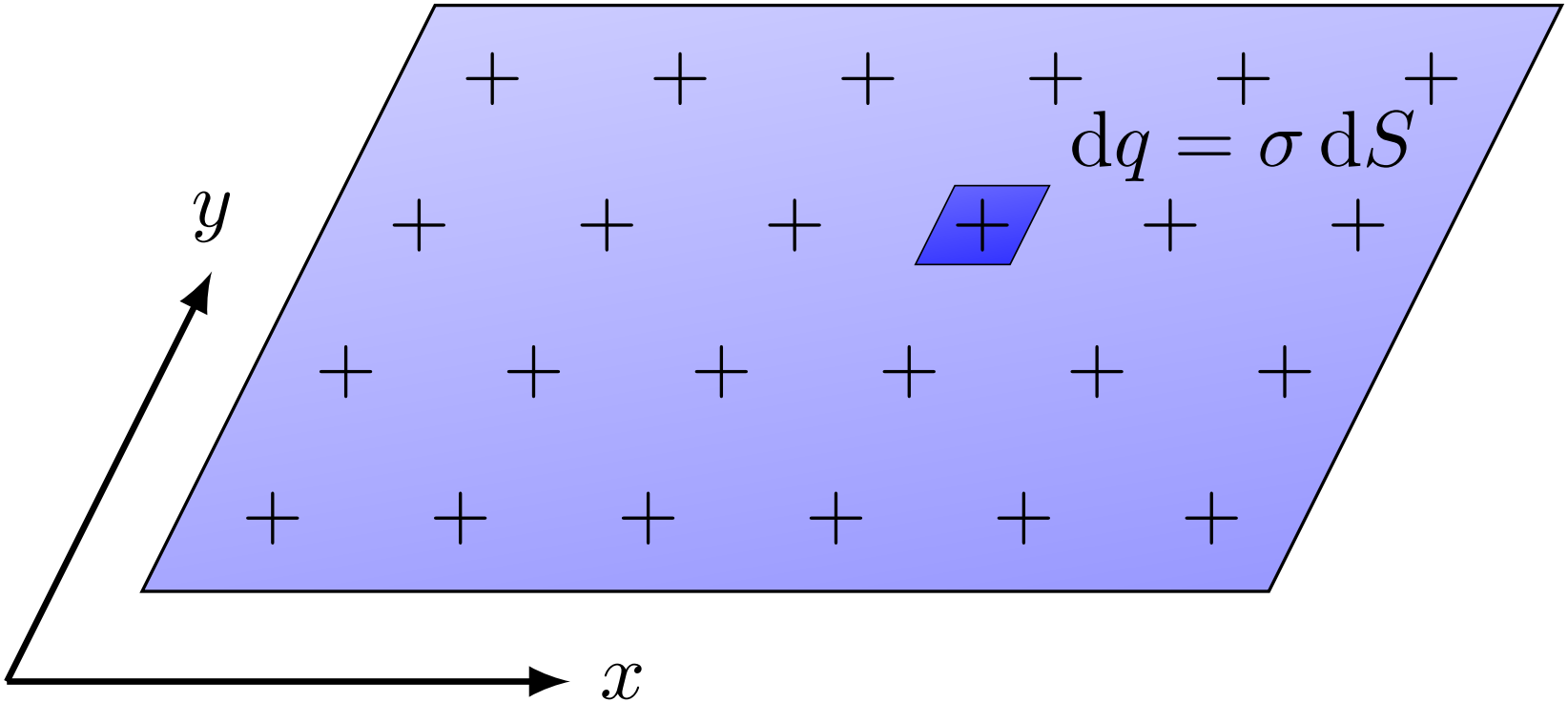
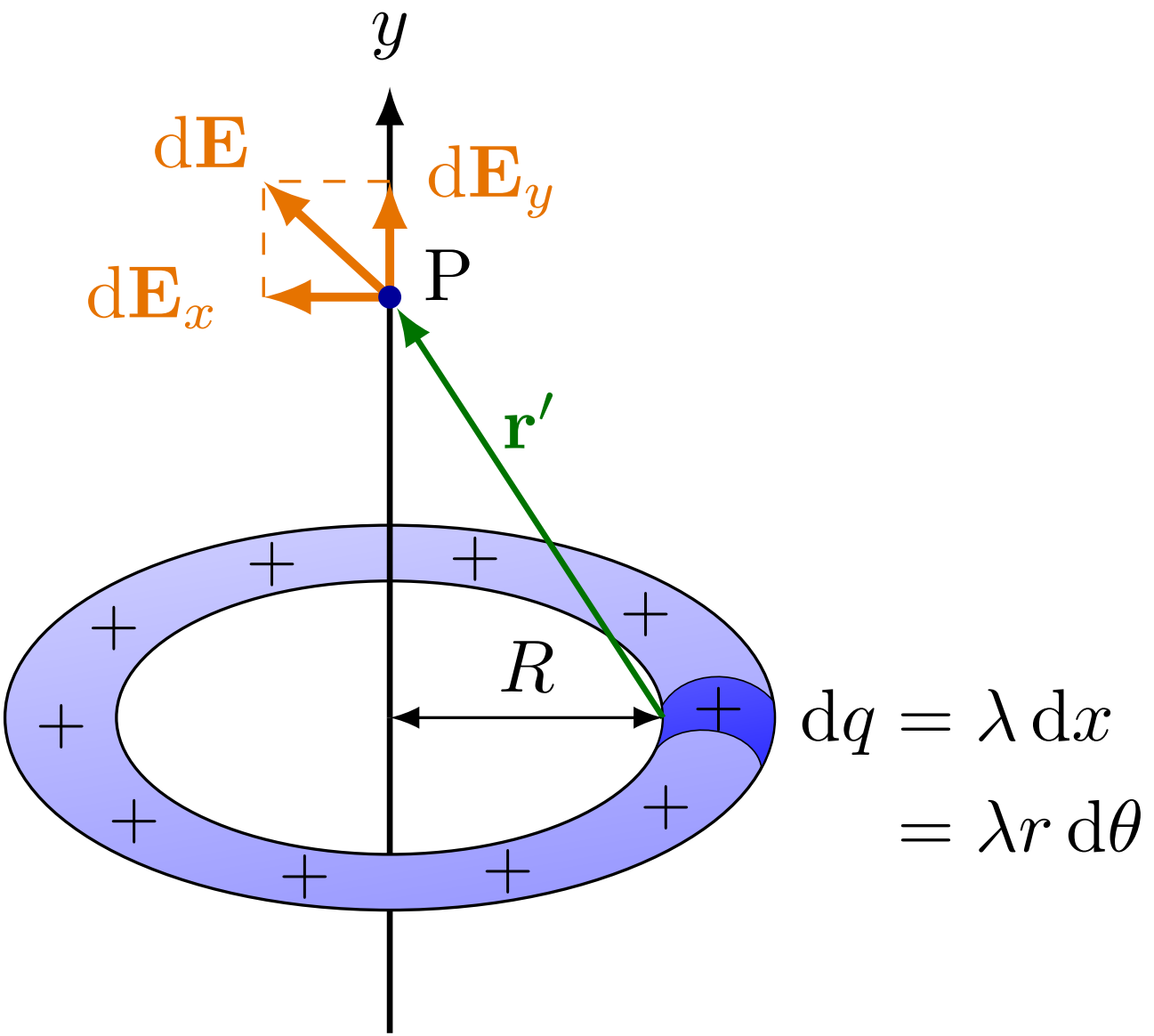
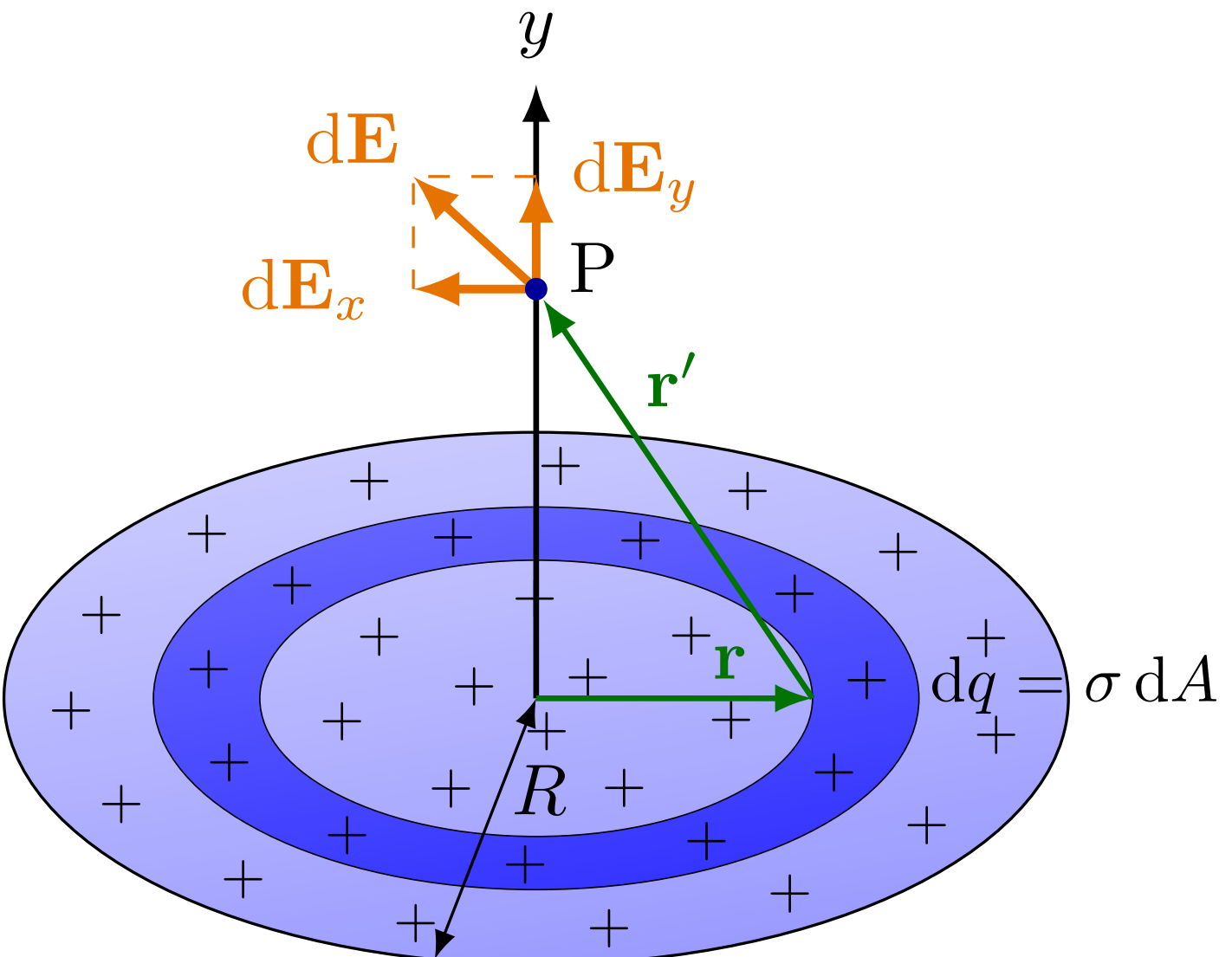
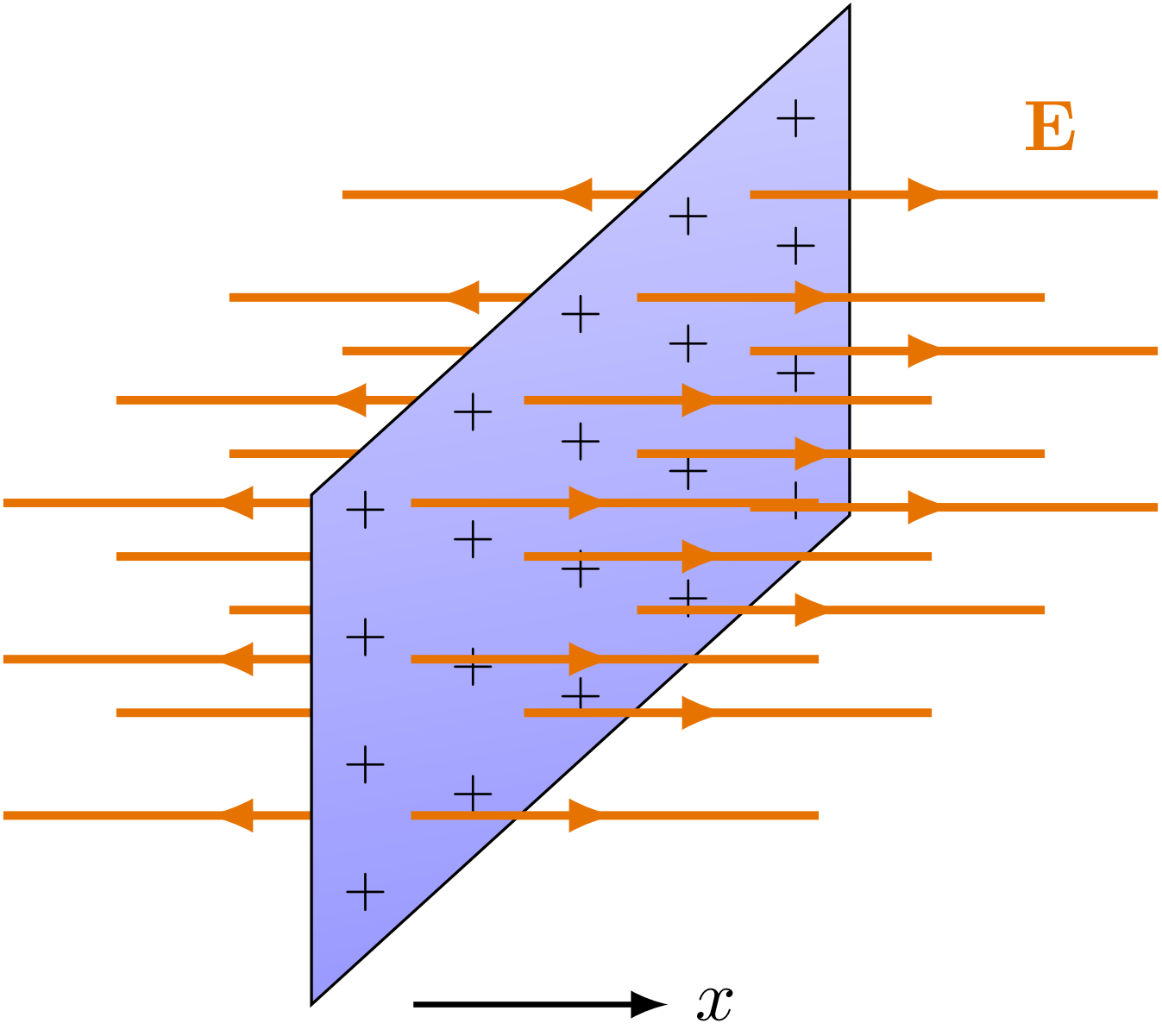
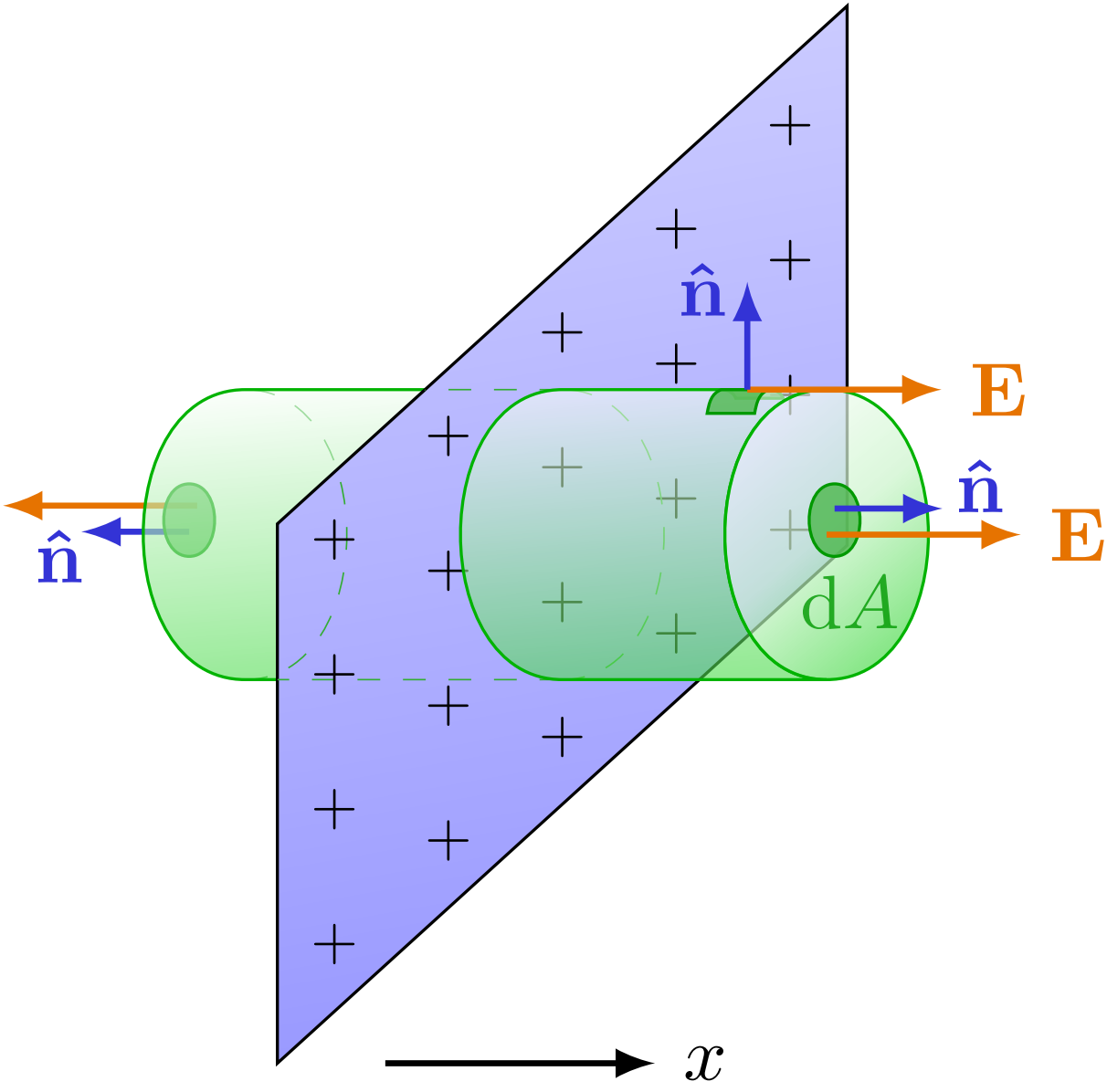
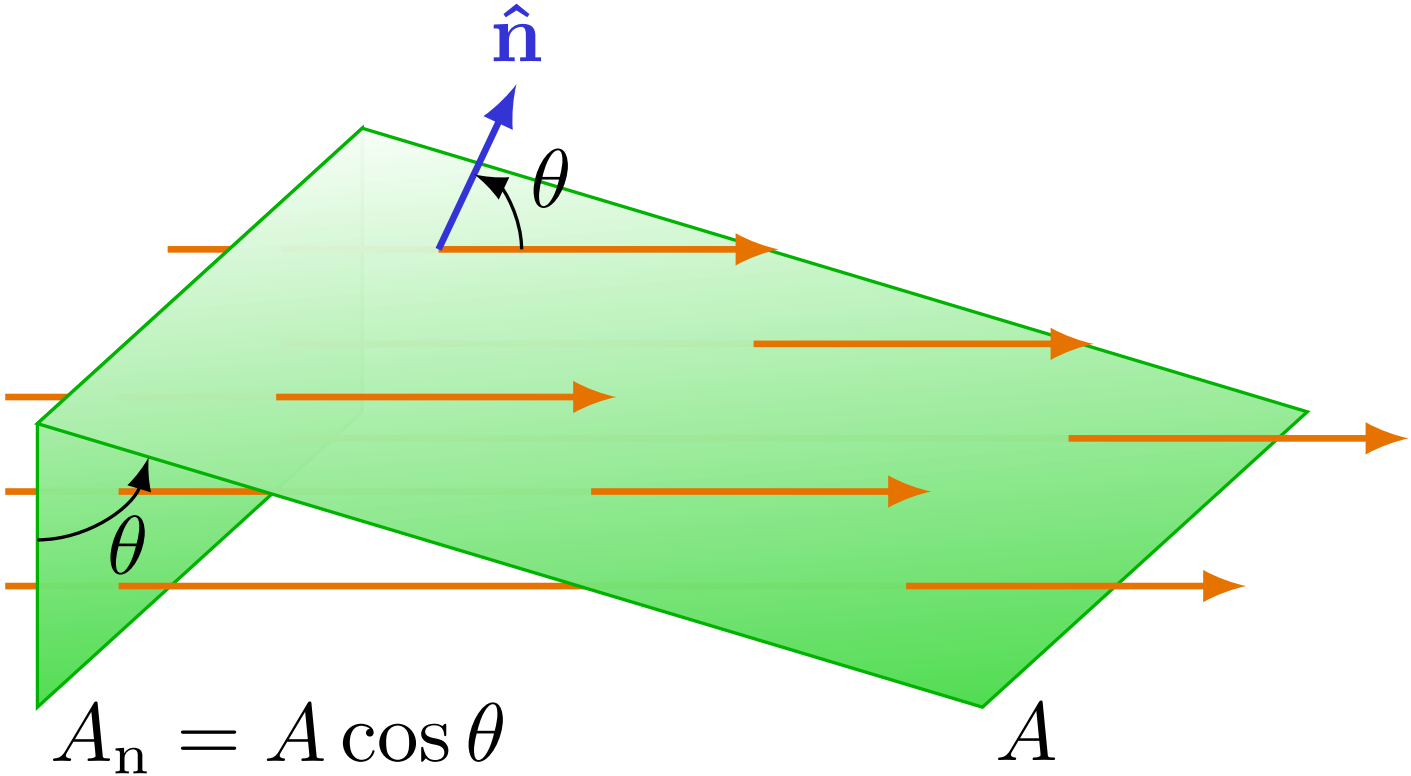

$\mathrm{d}q = \sigma\,\mathrm{d}S$
y
x


y
$\mathrm{d}\mathbf{E}$
$\mathrm{d}\mathbf{E}_y$
$\mathrm{d}\mathbf{E}_x$
P
$\mathbf{r}'$
R
$\mathrm{d}q = \lambda\,\mathrm{d}x$
$= \lambda r\,\mathrm{d}\theta$


y
$\mathrm{d}\mathbf{E}$
$\mathrm{d}\mathbf{E}_y$
$\mathrm{d}\mathbf{E}_x$
P
$\mathbf{r}'$
$\mathbf{r}$
$\mathrm{d}q = \sigma\,\mathrm{d}A$
R


E
x


n̂
E
n̂
E
n̂
dA
x


$\hat{\mathbf{n}}$
θ
θ
$A_{\mathrm{n}} = A\cos\theta$
A